

NOVA: Query-Specific Audio Grounding

Same audio x , opposite event orders



q_+ : dog followed by rooster

q_- : rooster followed by dog

Both windows are real.

The query determines the correct answer.

$x, q_{\pm}$



Core module: RelTwin

NOVA inference

Candidate generation
Official / SFT / SetPO

Evidence routing*
Keep / Drop scores

Boundary refinement
Radius + DP + gate

Final timestamp set
Query-aligned intervals

*Routing follows the evaluation configuration.

train

Audio-language model
Frozen backbone Trainable LoRA

(a) Answer competition

	$y(W_+)$	$y(W_-)$
q_+	$s_{++} \uparrow$	$s_{+-} \downarrow$
q_-	$s_{-+} \downarrow$	$s_{--} \uparrow$

s : mean token log-likelihood

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2} [\log p_+(+) + \log p_-(-)]$$

■ matched □ swapped

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}$$

Complete timestamp answers

$y(W_+)$	$y(W_-)$
[1.5, 11.75]	[14.75, 25.0]

(b) Query-specific outputs

